

Mahantesh Khetri

Ph.D. Candidate, Electrical and Computer Engineering | University of Virginia

[Laser Processing](#) | [Optical Systems](#) | [Semiconductor Metrology](#)

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Professional Summary

Ph.D. candidate in Electrical and Computer Engineering, expected December 2026, with hands-on experience in **laser micromachining, optical instrumentation and semiconductor metrology**. Develop process recipes, integrate optics and motion systems, and use Python and machine vision to turn measurements into repeatable experimental workflows.

Technical Skills

Laser Processing: Cutting, ablation, debonding, patterning and surface texturing; nanosecond, picosecond and CW systems; sapphire, silicon and SiC processing.

Optical Systems: Alignment, beam delivery, galvo scanners, F-theta optics, spectroscopy, ellipsometry, system integration and equipment troubleshooting.

Automation & Data: Galil motion control, EZCAD, Python, NumPy, pandas, SciPy, OpenCV, U-Net segmentation, vector scan paths and thermal modeling.

Materials & Metrology: PLD, thermal evaporation, high vacuum, SEM/EDS, TEM/STEM, Raman, fluorescence, LIBS, FTIR, UV-Vis and electrical characterization.

Research & Engineering Experience

Graduate Research Assistant / Doctoral Researcher

Aug 2022–Present

University of Virginia · Charlottesville, VA

- Develop laser cutting and micromachining processes for **sapphire, silicon and SiC**, alongside selective material removal, patterning and surface texturing.
- Optimize power, focus, scan speed, overlap and pass count for silicon/CIGS contact removal; demonstrated **single-pass CIGS debonding at 1,200 mm/s** scan speed.
- Convert electrode images into laser paths using OpenCV and U-Net segmentation, achieving **>98% pixel-level segmentation accuracy** on the reported detection task.
- Program **Galil-controlled stage motion** for repeated scans and hatch patterns; configure EZCAD and galvo workflows for selective processing.
- Align and troubleshoot laser, optical, scanning and cooling subsystems; document procedures and train users in equipment operation and laser safety.
- Develop multimodal optical instrumentation and semiconductor-interface measurement workflows; use **ellipsometry**, microscopy and spectroscopy to evaluate materials and process response.

Research Fellow

Dec 2019–Jul 2022

Indian Institute of Technology Delhi · New Delhi, India

- Operated PLD, high-vacuum and crystal-growth systems; correlated processing with microstructure and electrical/electrochemical properties using SEM, TEM, XRD, XPS and PPMS.

Additional responsibility: **PPMS Operator, January–July 2022.**

Earlier Research Experience

Research Student

Jul 2018–May 2019

Indian Institute of Technology Ropar · Rupnagar, India

- Fabricated Sb-doped SnO₂ transparent conducting oxide films using thermal evaporation and post-oxidation; evaluated structural, optical and electrical properties.

Research Intern, National Institute of Technology Warangal · Summer 2017

Selected Projects

Multimodal Spectroscopy Instrumentation

Developed a 355 nm Raman/fluorescence/LIBS platform using common excitation and collection optics, fiber-coupled spectroscopy and motorized positioning for co-localized measurements.

Semiconductor-Interface Metrology

Analyzed STEM-CL spectra registered to HAADF images of TiO_x/Al₂O₃/Si contacts; tested spectral reconstruction and interference correction to distinguish reproducible contrast from processing artifacts.

Low-Light Computational Imaging

Analyzed RAW sensor noise and conducted controlled restoration experiments using U-Net and Restormer across Sony Bayer and Fuji X-Trans data; separated variance stabilization from residual noise structure.

Education

Ph.D., Electrical and Computer Engineering, University of Virginia

Expected December 2026 · GPA: 3.9/4.0

M.Sc., Physics, Indian Institute of Technology Ropar

2019 · CGPA: 8.47/10

B.Sc., Computer Science, Physics and Mathematics, Rani Channamma University

2016 · 94.97% · University Second Rank; Gold Medal in Computer Science

Selected Publications & Patent

- [1] M. Khetri, P. K. Kanaujia, and M. C. Gupta. “Laser recycling of silver in bulk and nanoparticle form from silicon solar cells and deep learning for process automation,” *Renewable Energy*, vol. 254, 123684, 2025. doi: [10.1016/j.renene.2025.123684](https://doi.org/10.1016/j.renene.2025.123684)

- [2] M. Khetri, P. K. Kanaujia, and M. C. Gupta. “Laser removal of silicon solar cell backsheet while preserving tempered glass and silicon wafer,” *Solar Energy Materials and Solar Cells*, vol. 304, 114404, 2026. doi: [10.1016/j.solmat.2026.114404](https://doi.org/10.1016/j.solmat.2026.114404)

Co-inventor: Use of Lasers to Selectively Remove Materials from Substrates. Published U.S. patent application US20260008129A1 (2026).

Selected Honors

Wilson Bicentennial Graduate Fellowship, UVA (2025–2026) · First Place, Electrical Engineering Poster, UVERS (2025) · Second Place, Invited Speaker, UVERS (2025)